From: MEGE, Alexandre <alexandre.mege@airbus.com>
Sent: Friday, May 29, 2020 5:42 PM
To: lightweight-crypto
Cc: lwc-forum@list.nist.gov
Subject: ROUND 2 OFFICIAL COMMENT: HYENA

Dear all LWC members,
I hope you and your family are well in those difficult times.

It seems the round 2 version of Hyena is identical to the round 1 version ,and does not include the updated specification posted on April 27, 2019.
The round 2 version without the update is vulnerable to forgery.
The updated version from April 27,2019 is not vulnerable.

Source of vulnerability:
For a complete block the update of the variable Δ is Δ <= 2^t ⊙ Δ, with t = 2 if the last block of M is complete, or 3 if fractional. If a full block message is shortened by one block and the last block is made incomplete, then the same Δ is reused for the last block of the full message and the last block of the shortened message. This leads to potential forgery.

Detailed attack:
Considering a first known ciphertext message :
{C, T} = E(k, N, A, M)

with M = M0 || M1 || M2 and C = C0 || C1 ||C2, with M1 and M2 full blocks, and M0 any message of full blocks.

M1_l is the left half of M1
M2_l is the left half of M2

M1_r is the right half of M1
M2_r is the right half of M2

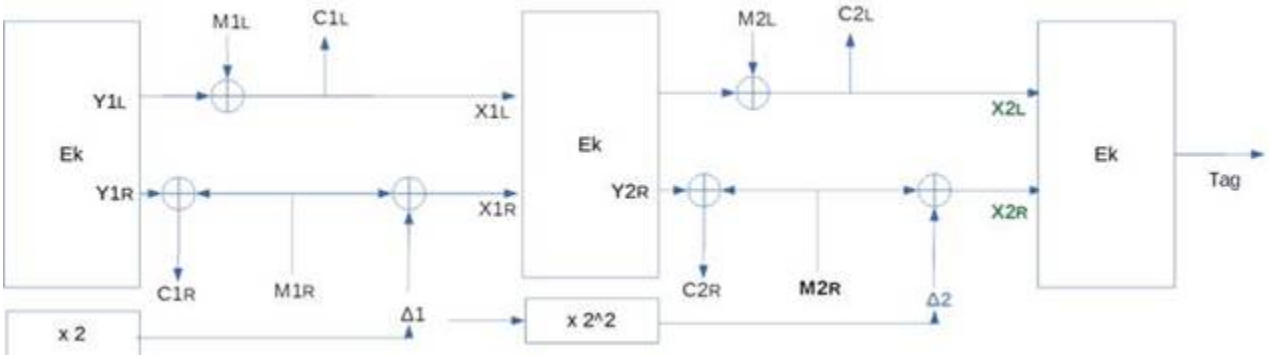
Let's define a message with

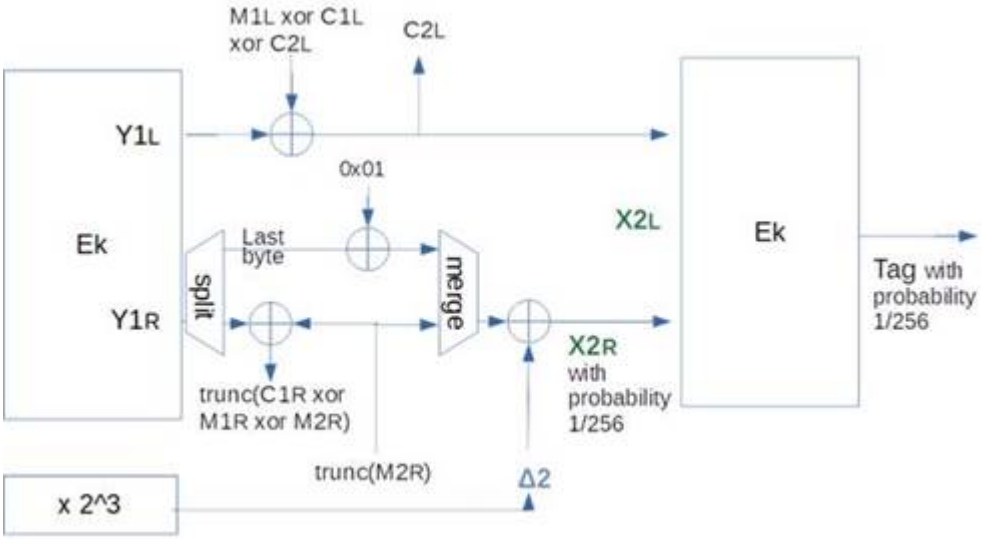
Mforge = M0|| (M1_l xor C1_l xor C2_l) || trunc(M2_r)
with trunc(x) the truncation the block x with last byte removed.

Then we can forge E(k, N, A, Mforge) with

{C0|| C2_l || trunc(M2_r xor M1_r xor C1_r) ,T}

The attack is successful with probability 1/256.
This happens when lastbyte(C1_R) xor 0x01 == lasbyte(C2_R).
This case can be checked offline to produce forged message that will pass tag check with probability 1.





Best regards,
Alexandre Mege

Example of forgery:

Key = 08831A952CA73EC940DB52ED64FF7601
Nonce = F03674BAF83E7C4280C6044A
PT = 00000000000000000000000000000089000000000000000000000000000000000
AD =
CT = 747C026790441425 FFCB95B5FE9A75 88 5D2833F76A3C739F AF4709777C5BE2A4
93C150F0BD5A544732396458D5CEB50F

Forge with

Key = 08831A952CA73EC940DB52ED64FF7601
Nonce = F03674BAF83E7C4280C6044A
PT = 29543190FA7867BA00000000000000
AD =
CT = 5D2833F76A3C739F FFCB95B5FE9A75 **93C150F0BD5A544732396458D5CEB50F**

From: lwc-forum@list.nist.gov on behalf of NILanjan Datta <nil.wid.frnds@gmail.com>
Sent: Saturday, May 30, 2020 5:09 AM
To: lwc-forum
Cc: lightweight-crypto
Subject: [lwc-forum] Re: ROUND 2 OFFICIAL COMMENT: HYENA
Attachments: main.pdf

Dear Alexandre,

Hope you are doing well. Thanks a lot for showing interest and analyzing our design.

We would like to clarify that we had already identified the mistake in Algorithm Proc_TXT [Line: 10] where the value 2^t should have been replaced by 3^t and informed it in the lwc-forum group mail dated April 29, 2019 (depending on the time zone) with the proposed modification. However, the updated version was not considered in the 2nd round submission due to the competition policy of NIST. Given that the submission proceeds to the next round, we will consider the same update. The attack that you have presented is on the initial version and has exploited the overlooked mistake.

We would also like to mention that we have analyzed the updated version and this version of HyENA (spec + security analysis + FPGA H/W implementation results) has recently been accepted in the ToSC Special Issue on Designs for the NIST Lightweight Standardisation Process. You can find the paper attached with the mail.

Please let us know, if you need any information from us.

Thanks and regards,
Designers, HyENA

On Saturday, 30 May 2020 03:12:40 UTC+5:30, alexandre.mege wrote:

> Dear all LWC members,
>
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